

Akylas Stratigakos

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Research Interests

My research develops and validates novel predictive and prescriptive analytics methods, leveraging machine learning, operations research, and statistics, to rapidly improve the planning and operations of low-carbon power and energy systems, reducing consumer costs and accelerating decarbonization.

Academic Appointments

- 08/2025 -** **Lecturer (Assistant Professor)**
UCL Energy Institute, University College London, UK
Energy Systems and Data Analytics
- 10/2023 - 07/2025** **Research Associate**
EEE Department, Imperial College London, UK

Education

- 2023** **Ph.D., Energy and Processes**
Mines Paris, PSL University, Sophia Antipolis, FR
Thesis: Towards the Prescriptive Analytics Paradigm for Energy Forecasting and Power System Optimization
- 2016** **Meng, Electrical and Computer Engineering**
University of Patras, Patras, GR
Specialization: Power Systems, Control

Visiting Positions

- 09/2022 - 01/2023** **Visiting Researcher**
OASYS Research Group, University of Malaga, ES

Publications

Journal Articles

- [1] **A. Stratigakos** and P. Andrianesis. “Learning Data-Driven Uncertainty Set Partitions for Robust and Adaptive Energy Forecasting with Missing Data”. In: *IEEE Transactions on Smart Grid* (2026). URL: <https://arxiv.org/abs/2503.20410>.
- [2] G. Van Caelenberg, **A. Stratigakos**, and E. Spyrou. “Constructing Deployment Scenarios for Reserve Deliverability via Adaptive Robust Optimization”. In: *Power Systems Computation Conference 2026* (2026). URL: <https://arxiv.org/abs/2603.16752>.
- [3] C. Bergmeir, F. De Nijs, E. Genov, A. Sriramulu, M. Abolghasemi, R. Bean, J. Betts, Q. Bui, N. T. Dinh, N. Einecke, R. Esmailbeigi, S. Ferraro, P. Galketiya, R. Glasgow, R. Godahewa, Y. Kang, S. Limmer, L. Magdalena, P. Montero-Manso, D. Peralta, Y. P. S. Kumar, A. Rosales-Pérez, J. Ruddick, **A. Stratigakos**, P. Stuckey, G. Tack, I. Triguero, and R. Yuan. “Predict+Optimize Problem in Renewable Energy Scheduling.” In: *IEEE Access* (2025), pp. 1–1. DOI: [10.1109/ACCESS.2025.3555393](https://doi.org/10.1109/ACCESS.2025.3555393).
- [4] **A. Stratigakos**, S. Pineda, and J. M. Morales. “Decision-focused linear pooling for probabilistic forecast combination”. In: *International Journal of Forecasting* (2024). ISSN: 0169-2070. DOI: <https://doi.org/10.1016/j.ijforecast.2024.11.006>. URL: <https://www.sciencedirect.com/science/article/pii/S0169207024001213>.

- [5] **A. Stratigakos**, S. Pineda, J. M. Morales, and G. Kariniotakis. “Interpretable Machine Learning for DC Optimal Power Flow With Feasibility Guarantees”. In: *IEEE Transactions on Power Systems* 39.3 (2024), pp. 5126–5137. DOI: [10.1109/TPWRS.2023.3333165](https://doi.org/10.1109/TPWRS.2023.3333165).
- [6] **A. Stratigakos**, P. Andrianesis, A. Michiorri, and G. Kariniotakis. “Towards Resilient Energy Forecasting: A Robust Optimization Approach”. In: *IEEE Transactions on Smart Grid* (2023), pp. 1–1. DOI: [10.1109/TSG.2023.3272379](https://doi.org/10.1109/TSG.2023.3272379).
- [7] K. Krommydas, C. Dikaiakos, G. Papaioannou, and **A. Stratigakos**. “Flexibility study of the Greek power system using a stochastic programming approach for estimating reserve requirements”. In: *Electric Power Systems Research* 213 (2022), p. 108620.
- [8] K. F. Krommydas, **A. C. Stratigakos**, C. N. Dikaiakos, G. P. Papaioannou, M. G. Jones, and G. C. McLoughlin. “A Novel Modular Mobile Power Flow Controller for Real-Time Congestion Management Tested on a 150kV Transmission System”. In: *IEEE Access* 10 (2022), pp. 96414–96426. DOI: [10.1109/ACCESS.2022.3205589](https://doi.org/10.1109/ACCESS.2022.3205589).
- [9] **A. Stratigakos**, S. Camal, A. Michiorri, and G. Kariniotakis. “Prescriptive trees for integrated forecasting and optimization applied in trading of renewable energy”. In: *IEEE Transactions on Power Systems* 37.6 (2022), pp. 4696–4708.
- [10] **A. Stratigakos**, A. Bachourmis, V. Vita, and E. Zafiroopoulos. “Short-term net load forecasting with singular spectrum analysis and LSTM neural networks”. In: *Energies* 14.14 (2021), p. 4107.
- [11] G. P. Papaioannou, C. Dikaiakos, **A. C. Stratigakos**, P. C. Papageorgiou, and K. F. Krommydas. “Testing the efficiency of electricity markets using a new composite measure based on nonlinear TS Tools”. In: *Energies* 12.4 (2019), p. 618.

Conference Papers

- [12] K. Krommydas, **A. Stratigakos**, E. Chassioti, and I. Moraitis. “A Two-Stage Stochastic Unit-Commitment Formulation for Evaluating the Impact of Battery Energy Storage Systems on Reserve Requirements”. In: *14th Mediterranean Conference on Power Generation Transmission, Distribution and Energy Conversion*. IEEE. Athens, Greece, Dec. 2024.
- [13] K. F. Krommydas, **A. C. Stratigakos**, E. Chassioti, and I. Moraitis. “A two-stage stochastic unit-commitment formulation for evaluating the impact of battery energy storage systems on reserve requirements”. In: *IET Conference Proceedings CP904*. Vol. 2024. 29. IET. 2024, pp. 684–689.
- [14] M. Kühnau, **A. Stratigakos**, S. Camal, S. Chevalier, and G. Kariniotakis. “Resilient Feature-driven Trading of Renewable Energy with Missing Data”. In: *2023 IEEE Power & Energy Society Innovative Smart Grid Technologies conference (ISGT)*. 2023. URL: <https://hal.science/hal-04104548>.
- [15] **A. Stratigakos**, D. van der Meer, S. Camal, and G. Kariniotakis. “End-to-end Learning for Hierarchical Forecasting of Renewable Energy Production with Missing Values”. In: *2022 17th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*. 2022, pp. 1–6. DOI: [10.1109/PMAPS53380.2022.9810610](https://doi.org/10.1109/PMAPS53380.2022.9810610).
- [16] **A. Stratigakos**, A. Michiorri, and G. Kariniotakis. “A Value-Oriented Price Forecasting Approach to Optimize Trading of Renewable Generation”. In: *2021 IEEE Madrid PowerTech*. 2021, pp. 1–6. DOI: [10.1109/PowerTech46648.2021.9494832](https://doi.org/10.1109/PowerTech46648.2021.9494832).
- [17] K. F. Krommydas, **A. C. Stratigakos**, C. Dikaiakos, G. P. Papaioannou, E. Zafiroopoulos, and L. Ekonomou. “An improved flexibility metric based on kernel density estimators applied on the Greek power system”. In: *Flexitranstore*. Vol. 610. Springer, 2020, pp. 35–46.
- [18] **A. C. Stratigakos**, K. F. Krommydas, P. C. Papageorgiou, C. Dikaiakos, and G. P. Papaioannou. “A Suitable Flexibility Assessment Approach for the Pre-Screening Phase of Power System Planning Applied on the Greek Power System”. In: *IEEE EUROCON 2019 -18th International Conference on Smart Technologies*. 2019, pp. 1–6.

Preprints

- [19] **A. Stratigakos**, H. Wen, E. Spyrou, and P. Pinson. “Decision-calibrated prediction sets for robust power system operations”. In: (2026). working paper or preprint. arXiv: [2606.02081 \[math.OC\]](https://arxiv.org/abs/2606.02081). URL: <https://arxiv.org/abs/2606.02081>.
- [20] J. G. D. la Varga, S. Pineda, **A. Stratigakos**, and J. M. Morales. “End-to-End Pseudo-Measurement Learning for State Estimation under Limited Observability”. In: (2026). working paper or preprint. arXiv: [2605.23525 \[math.OC\]](https://arxiv.org/abs/2605.23525). URL: <https://arxiv.org/abs/2605.23525>.

Thesis

- [21] **A. Stratigakos**. “Towards the Prescriptive Analytics Paradigm for Energy Forecasting and Power System Optimization”. Theses. Université Paris sciences et lettres, July 2023. URL: <https://pastel.hal.science/tel-04250526>.

Teaching Experience

- 2025 - 2026** **Energy Data Analytics in the Built Environment (Module Lead)**
Energy Systems and Data Analytics MSc (15 credits)
- Designed and led a core MSc module (>60 students) on data analytics methods to analyze, predict, and optimize building energy demand and use.

Industry Experience

- 09/2017 - 05/2020** **Research and Development Engineer**
Independent Power Transmission Operator (IPTO), Athens, GR
- Developed tools to assess system flexibility under high renewable penetration.
 - Coordinated successfully installing an innovative power flow controller.
 - Contribution to Horizon 2020 projects.
- 01/2022 - 06/2023** **Data Scientist - Volunteer**
GIVMED, N.G.O., Athens, GR
- Redistributing surplus medical supplies to socially vulnerable groups and addressing healthcare inequalities through data analytics.

Grants & Awards

- RESEARCH GRANTS** LLMs4Energy: Large Language Models for Energy Systems Modelling (PI, £7.5k)
- AWARDS** Outstanding Reviewer IEEE Transactions on Sustainable Energy (2025)
Best PhD Thesis Award, Think Smartgrids Association (2024)
Erasmus+ Mobility Grant (2023)
Best Student Presentation & Travel Grant Award, 42nd International Symposium on Forecasting (2022)

Supervision

- PHD** **Guillaume Van Caelenberg**, Imperial College London, 2023 - 2026 (expected).
Reserve Deliverability in Co-Optimized Electricity Markets (primary supervisor: Dr Elina Spyrou).
Steven Sperazza, University College London, 2025 - 2029 (expected). *Supporting policymaking with statistical emulation of complex models* (primary supervisor: Dr Erica Thompson).
- MSc** 8 students supervised
- BSc/MENG** 2 students supervised

Leadership and Service

REVIEWER	<i>Journals:</i> IEEE Transactions on Smart Grids/ Power Systems/ Sustainable Energy; Electric Power Systems Research. <i>Conferences:</i> IEEE PES PowerTech; International Conference on Probabilistic Methods Applied to Power Systems (PMAPS); Power Systems Computation Conference (PSCC).
ORGANIZATION	Workshop: Global Power System Transformation Consortium AI Workshop, at ICMS, Edinburgh, 19-23/02/2024 (<i>co-organizer, session chair</i>) Invited Session: Value-oriented Forecasting, 44th International Symposium on Forecasting, Dijon, France (<i>organizer, chair</i>). Seminar Series: Inaugural event of the CIGRE UK London Region Universities Hub, at Imperial College London, London, 06/05/2025 (<i>co-organizer</i>).
SERVICE	CIGRE UK University Hub London Region (<i>Vice Chair, 2025-2026</i>). 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys 2026) (<i>technical program committee member</i>)

Additional Information

LANGUAGES	Greek (native), English (proficient), German, French (basic)
SOFTWARE	Python, MATLAB, R, Julia
MEMBERSHIPS	IEEE, CIGRE, ESIG, International Institute of Forecasters